Tire Development Instruction Package
NV56 – Base All Season – 19 Square

255/45R19 104W XL

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TECHNICAL SPECIFICATION #: **TS-1188211**
REVISON #: 01

DOCUMENT HISTORY

Rev	Date	Description	Author
01	18APR2018	Initial Release	Guzman

Table of Contents

1. Scope and Objective

This document provides attributes for performance and durability validation of Tesla part number:

1188211-XX-X 255/45R19 104 W XL

Any information not contained in this specification, but listed separately on other Tesla Motors official documentation (such as part drawing, etc.) shall also be satisfied.

2. General Specification

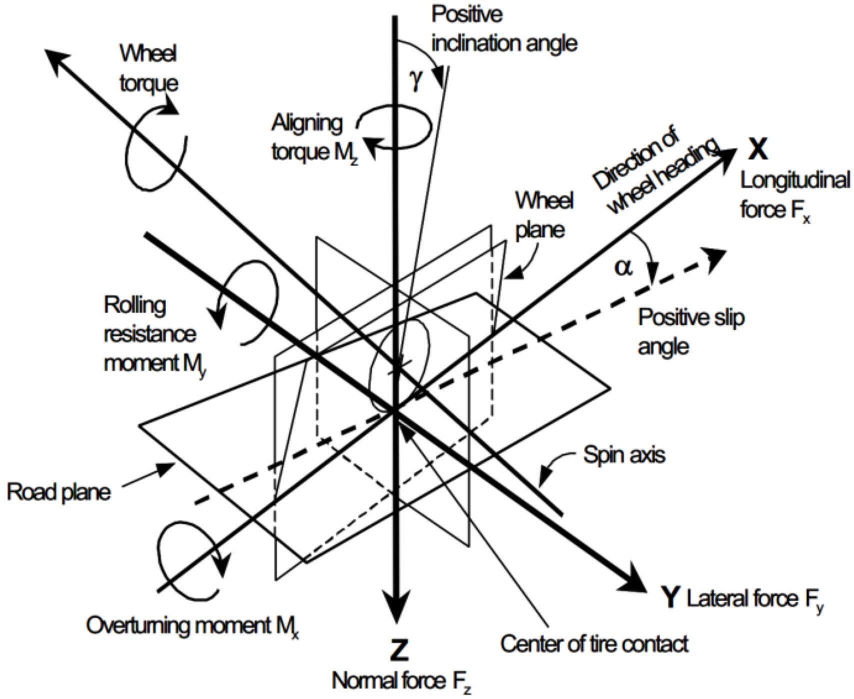
2.1. Terms and Definitions

Throughout this specification, various terms are used and their definitions must be understood for proper interpretation of this document. These terms and definitions are listed below.

Term	Definition
F_Y	Force in the Y direction, also called Lateral Force
F_X	Force in the X direction, also called Longitudinal Force
M_Z	Moment about the Z axis
SA	Slip Angle is the angle created between the path that the tire is traveling and the direction the tire is pointing.
SR	Slip Ratio is the ratio of the longitudinal travel of the wheel divided by the travel of the tire.
SLR	Static Load Radius is the radius of the tire measured from the center of the tire to the ground plane at a specific vertical load while the vehicle is no moving.
ERR	Equivalent Rolling Radius is the radius of the tire measured from the center of the tire to the ground plane while rotating at a certain speed and at a specific vertical load. Typically ERR is back calculated from revolutions per distance traveled.
PRAT	Plysteer Residual Aligning Torque
PRCF	Plysteer Residual Cornering Force

2.2. Coordinate System Definition

All forces and moments follow the SAE tire axis system as defined in SAE J2047.



2.3. Vehicle Information

Summary Vehicle Information relevant to tire application is listed below.

Item Name	Units	Data Point Front	Data Point Rear	Note
Applied Tire Pressure	kPa	290	290	Tuning Item
Tire Load - Curb	kg	525	550	-
Tire Load – Performance	kg	560	580	2 Up
Tire Load - GVW	kg	600	725	-
Tire Load – GVW + Towing	kg	600	815	-
Top Speed	kph	225		-
Tire Life	km	55K	55K	Rotation Allowed
Peak Power	W	-		-
Application Rim Size	inch	19x9.5J	19x9.5J	TBC
Acoustic Foam	-	Yes	Yes	Density Tuning Item
Sensor on Tire	-	Required		BLE TPMS
Application Camber	Deg	-0.5	-1.0	Tuning Item
Application Toe	Deg	0	0.2IN	Tuning Item
Tire Size	-	255/45R19	255/45R19	
Load Index	-	104	104	
Speed Rating	-	W	W	
Tesla Marking	-	T0	T0	Engraved
Maximum Factory New Tire Age	Weeks	26	26	
Maximum Service New Tire Age	Weeks	52	52	

2.4. Tire Construction Reporting

Each unique tire constriction shall have a corresponding Tire Construction Sheet. Design Intent, Tread Details, and Construction Details from template below to be provided before each Submission as defined in Section 3. Additional data to be provided for Preferred Constructions as defined in Section 3.

Table 1: Tire Construction Sheet. Template can be provided by Tesla Chassis Engineering

		Submission # 1		
		1	2	3
Constr. Code		**123456A**	**123456B**	**123456C**
Tire supplier's description of design intent for each construction		Regular	Lowest RRC	
Tread				
Tread Compound		123456	←	←
Under-Tread Compound		789123	←	←
Construction Details				
Overlay material		140/1	←	←
Overlay design (center, shoulder, angle)		Classic	←	←
Belt material & type		Steel - Hard Rubber 4x30	Steel - Soft Rubber 4x30	←
Top belt cured angle		+/- 26°	←	←
Sidewall Material		2 ply	←	←
Body ply material		Polyester	←	←
Bead matrix/design		2 piece	←	←
Apex/bead filler material		Hard/Hard	←	Soft/Soft
Apex/bead filler height (mm)		60	←	←
Tire Dimensions & Mechanics				
Tire mass (kgs)				
Outer Diameter (mm)				
Section Width (mm)				
Tread depth - center (mm)			←	←
Tread depth - shoulder (mm)			←	←
Footprint (area, shape, width ratio)		*Picture w/ void ratio within footprint*		
RPK		*report in 2D table as specified in TDIP*		
DLR		*report in 2D table as specified in TDIP*		
SLR (70% of Load) (mm)	Front Axle			
SLR (100% of Load) (mm)	Front Axle			
ERR (100% Load) (mm)	Front Axle			
SLR (70% of Load) (mm)	Rear Axle			
SLR (100% of Load) (mm)	Rear Axle			
ERR (100% Load) (mm)	Rear Axle			
Rolling Resistance				
RRc - SAEJ2452 Coefficients	α			
	β			
	a			
	b			
	c			
RRc - SAEJ2452 (kg/ton) (@ TDIP Conditions)				
Force and Moment (MF 6.2 with SWIFT)				
Cornering Stiffness (Normalized Fy/SA)		*.tir file Report Criteria from TDIP*		
Tractive Force (Normalized Fx/SA)				
Aligning Torque Stiffness (Normalized Mz/SA)				
XYZ Stiffness (N/mm)				
Dynamic Stiffness (Relaxation Length, m/SA)				
PRAT (Nm) *Measured at PPAP				
Noise Vibration and Harshness				
Max Sound Pressure Level (dba at frequency)				
Max Structural Vibration (g at frequency)				

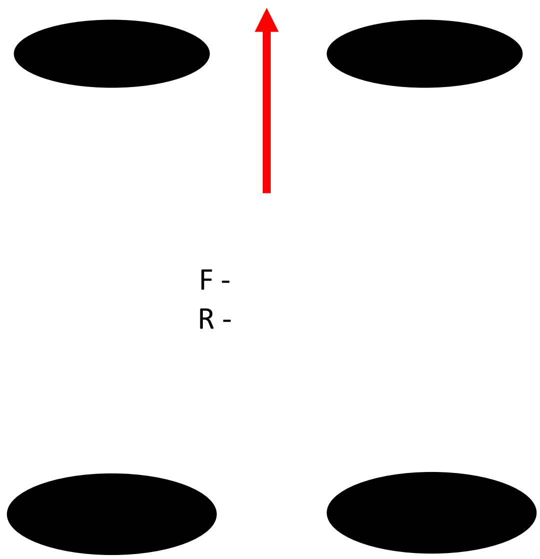
2.5. Tire Profile Reporting

Each preferred constriction as defined in Section 3, shall have a corresponding Tire Profile using analysis tools with design intent constructions and parameters.

	Vertical Load	Centrifugated Speed	Toe	Camber
	kg	kph	Deg	deg
Loaded Front	560	100	0.05 OUT	-0.5
Loaded Rear	580	100	0.20 IN	-1.0
Inflated F	0	0	0	0
Inflated R	0	0	0	0

2.6. Tread Pressure Distribution Reporting

Each preferred constriction as defined in Section 3, shall have a corresponding Tread Pressure Distribution Reporting using analysis tools with design intent constructions and parameters.



		Vertical Load	Lateral Load	Longitudinal Load	Pressure	Speed	Slip Angle	Slip Ratio	Camber Angle
		kg	N	N	kPa	kph	Deg	%	Deg
Condition 1	**Front**								
	Rear								
Condition 2	**Front**								
	Rear								
Condition 3	**Front**								
	Rear								

Conditions to be determined during development.

2.7. Virtual Tire Reporting

Each proposed construction shall have a virtual tire model submitted to Tesla Engineering before constructions are built. Virtual submission shall satisfy the format for Equation 2.6.1 for a given Normal Load. Normal load sensitivities for each Virtual Coefficient shall be provided through second order polynomial curve fits for each coefficient. Target values for each coefficient shall be set by Tesla Engineering.

$$F_y = D * \sin[C\, arctan\{B\alpha - E\,(B\alpha - arctan(B\alpha))\}] \tag{2.6.1}$$

Table 2: Pacejka MF simplified coefficients

Name	Coefficient	Unit	Equivalent	Requirement	Target
Lateral Force	F_y	N	F_y	Output	
Slip Angle	α	deg	α	Variable	
Cornering Stiffness	BCD	$\frac{N}{deg}$	C_α	Virtual	4.2.1
Peak Cornering	D	-	µ	Virtual	4.2.1
Shape Factor	C	-	*	Virtual	-
Stiffness Factor	B	-	$\frac{C\alpha}{C * D}$	Virtual	4.2.1
Curvature Factor	E	-	**	Virtual	-
Vertical Stiffness	K_Z	$\frac{N}{mm}$	-	Virtual	4.2.4
Lateral Stiffness	K_Y	$\frac{N}{mm}$	-	Virtual	4.2.5

* - Shape Factor [C] – usually constant function of F_Z.

** - Curvature Factor [E] – usually a second order polynomial function of F_Z.

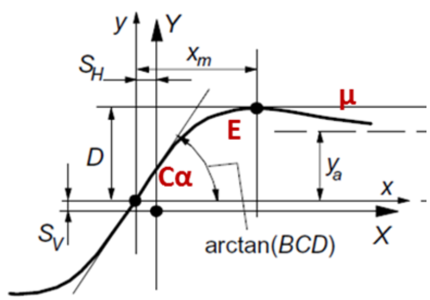


Figure 1: Proposed Virtual Output, graphical representation.

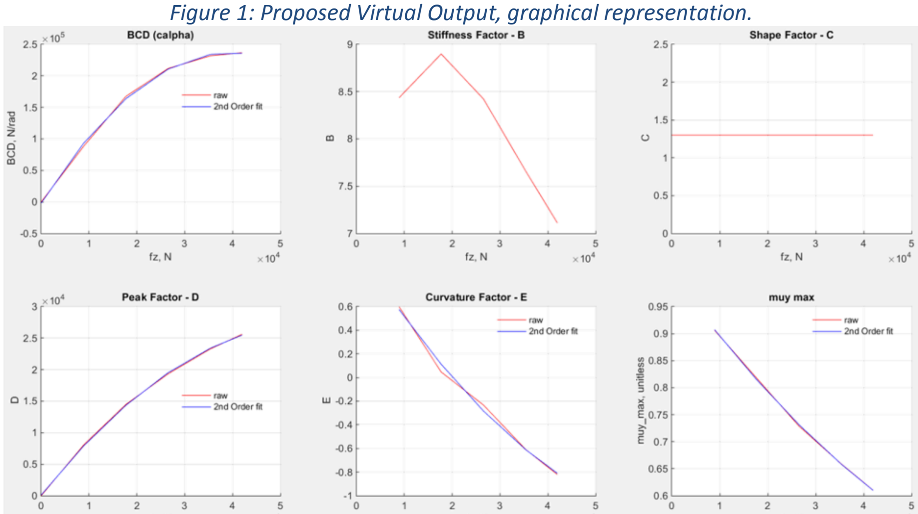


Figure 2: Normal Load Sensitivities for each Coefficient

3. Confirmation Item Summary

S# (Submission No.) tests are intended to develop the subjective rating correlation to various design engineering decisions. There may be multiple submissions to allow for iterations in component design.

PV (Production Validation) tests shall be completed using initial parts from production tooling and process and are intended to identify relationships between design and process characteristics.

IP (In-Process) tests shall be completed with production parts on an ongoing basis. Supplier shall be responsible for creating and maintaining Control Plan documentation to establish a basis for continuous improvement.

Results of all reports shall be presented in DVP&R and other more detailed format including pictures of test setups, actual test parts, and detailed tables of observations.

The following table summarizes the minimum requirements for DV and PV testing. Supplier shall propose additional tests where applicable to further guarantee life and performance for the entirety of the vehicle life.

Item	Item Name	Test Requirement		
4.1 Dimensional		S#	PV	IP
4.1.1	Width	A	•	Control Plan
4.1.2	Overall Diameter	A	•	Control Plan
4.1.3	Mass	A	•	Control Plan
4.1.4	Moment of Inertia	B	•	Control Plan
4.1.5	Static Loaded Radius	A	•	Control Plan
4.1.6	Dynamic Loaded Radius	C	•	Control Plan
4.1.7	Revolutions per Mile	C	•	Control Plan
4.2 Force and Moment (SWIFT MF6.2)				
4.2.1	Cornering Stiffness (Normalized F_Y/SA)	A	•	Control Plan
4.2.2	Aligning Torque Stiffness (Normalized M_Z/SA)	A	•	Control Plan
4.2.3	Tractive Force (Normalized F_X/SA)	B	•	Control Plan
4.2.4	Vertical Stiffness (N/mm)	A	•	Control Plan
4.2.5	Lateral Stiffness (N/mm)	A	•	Control Plan
4.2.6	Plysteer Residual Aligning Torque	C	•	Control Plan
4.2.7	Plysteer Residual Cornering Force	C	•	Control Plan
4.2.8	Vertical Damping	C	•	Control Plan
4.2.9	Dynamic Stiffness, Relaxation Length (m/SA)	A	•	Control Plan
4.3 Rolling Resistance				
4.3.1	RR_C (SAE J2452 method)	A	•	Control Plan
4.4 Noise Vibration & Harshness				
4.4.1	Max Sound Pressure Level	B	•	Control Plan
4.4.2	Max Structural Vibration (g at frequency)	B	•	Control Plan
4.5 Tread-wear				
4.5.1	Fastest wearing groove (mm/10,000 miles)	C	•	Control Plan
4.5.2	Average wear rate (mm/10,000 miles)	C	•	Control Plan
4.5.3	Irregular wear rating (mm/10,000 miles)	C	•	Control Plan
4.6 Subjective Handling				
4.6.1	Complete Table	•	•	Control Plan
4.7 Uniformity				
4.7.1	Conicity	C	•	Control Plan
4.7.2	Radial RunOut	C	•	Control Plan
4.7.3	Radial Force Peak to Peak	C	•	Control Plan
4.7.4	Radial First Harmonic	C	•	Control Plan
4.7.5	Lateral Force Peak to Peak	C	•	Control Plan
4.7.6	Lateral RunOut	C	•	Control Plan

A – Test to be completed for each construction, before submission. Virtual or simulation data OK.
B – Test to be completed for preferred construction, after submission
C – Test to be completed for preferred construction, after final submission
• – Test to be completed for each construction, during submission.

4. Confirmation Items

Tested components are to be of random sampling and not repeat the same tests, unless otherwise instructed by Tesla Engineering.

4.1. Dimensional

Item #	Item Name	Pressure (kPa)	Unit		Data Front	Data Rear
4.1.1	Overall Width	290	mm		**252 ± 3**	**252 ± 3**
4.1.2	Overall Diameter	290	mm		**719 ± 3**	**719 ± 3**
4.1.3	Mass	-	kg		**10.8 Max**	**10.8 Max**
4.1.4	Moment of Inertia	-	$kg*m^2$	I_{XX}	TBC	TBC
		-		I_{YY}	TBC	TBC

4.1.5. Static Loaded Radius

SLR (mm)		Pressure (kPa)		
Mass (kg)		**276**	**290**	**310**
Curb Front	**525**	340 ± 5	340 ± 5	340 ± 5
Curb Rear	**550**	340 ± 5	340 ± 5	340 ± 5
Greatest Vehicle Axle	**725**	335 ± 5	335 ± 5	335 ± 5

4.1.6. Dynamic Loaded Radius

Report the dynamic loaded radius for the following speeds at the application tire pressure:

DLR (mm)		Speed (kph)				
Mass (kg)		15	80	130	195	240
Performance Front	**560**					
Performance Rear	**580**					
Greatest Vehicle Axle	**725**					

4.1.7. Revolutions per Mile

Report revolutions per mile for the following speeds at the application tire pressure:

rev/km		Speed (kph)				
Mass (kg)		15	80	130	195	240
Performance Front	**560**					
Performance Rear	**580**					
Greatest Vehicle Axle	**725**					

$$ERR = \frac{distance\ traveled}{revolutions * 2\pi}$$

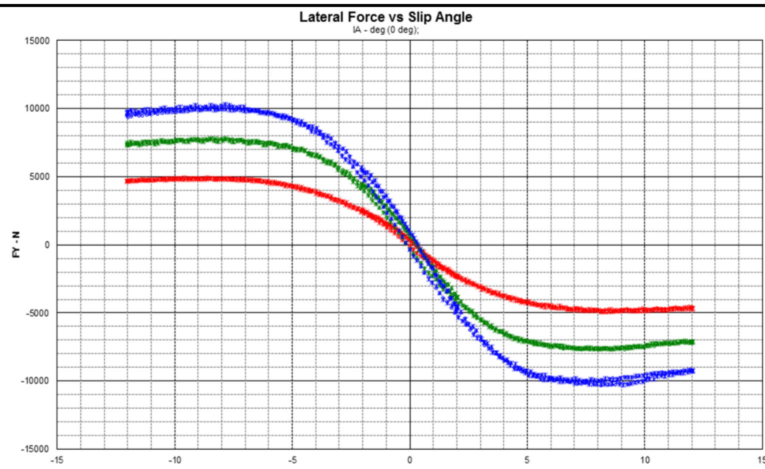
$$ERR = \frac{1\ km}{2\pi * 500\ revs} = .318\ m = 318\ mm$$

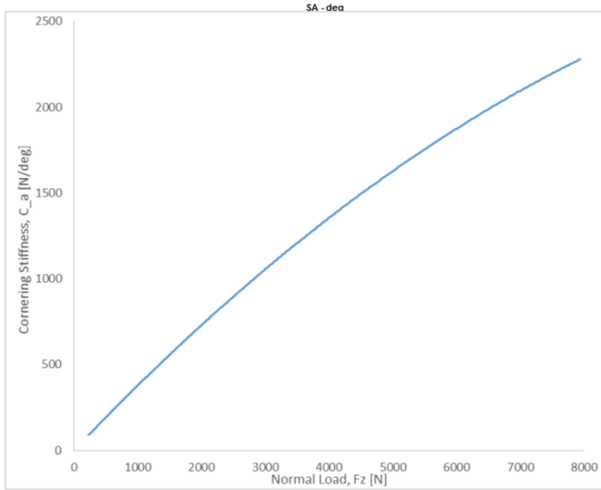
4.2. Force Moment

As measured by a force and moment, the output of the data should be **in MF6.2 format with SWIFT**, and adequate to create the below characterization plots. Data analysis will be performed by Tesla Motors Inc.

Load values, pressure, and resolution should be in accordance with standards and practices at the Supplier's test facility for passenger car/light truck. Collected F&M data will be available in the .tir file for each dimension.

4.2.1. Normalized Cornering Stiffness

<table>
<tr><td>Description</td><td>Cornering Coefficient is the force in the Y direction divided by the Slip Angle and normalized by vertical load.</td></tr>
<tr><td>Set Up</td><td>The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.</td></tr>
<tr><td>Image</td><td>




</td></tr>
<tr><td>Criteria</td><td>
<table>
<tr><th>Attribute</th><th>Parameters</th><th>Target</th><th>Units</th></tr>
<tr><td>Cornering Stiffness Coefficient</td><td>SA = 0
Fz = 0</td><td>.35 MIN</td><td>(N/deg)/N</td></tr>
<tr><td>Cornering Stiffness</td><td>Fz = 5.5 kN</td><td>1700 ± 5%</td><td>N/deg</td></tr>
<tr><td>Cornering Stiffness Coefficient OffCenter</td><td>SA = 0
Fz = 5.7 kN</td><td>TBD</td><td>(N/deg)/N</td></tr>
<tr><td>Normalized Lateral Force</td><td>Fz = 5.7 kN</td><td>TBD</td><td>-</td></tr>
</table>
Report Comparison to Reference Tire in 4.6
</td></tr>
</table>

4.2.2. Normalized Aligning Torque

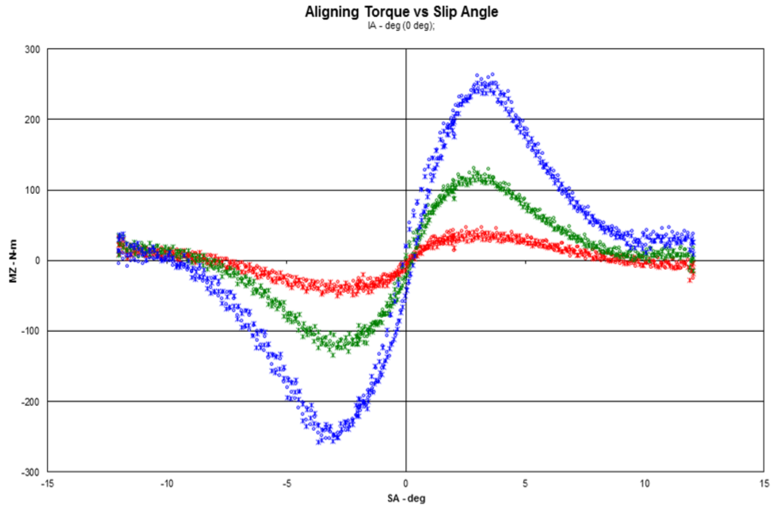
Description

Aligning torque stiffness is the moment about the z axis divided by the Slip Angle and normalized by vertical load.

Set Up

The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.

Image



Criteria

Attribute	Parameters	Target	Units
Aligning Torque Stiffness	Fz = 5.5 kN	90 ± 10%	Nm/deg
Pneumatic Trail	Fz = 5.7 kN SA @ $F_{Y,\ MAX}$	TBD	m
Peak Aligning Moment	Fz = 5.7 kN	TBD	Nm

Report comparison to Reference Tire in 4.6

4.2.3. Normalized Tractive Force

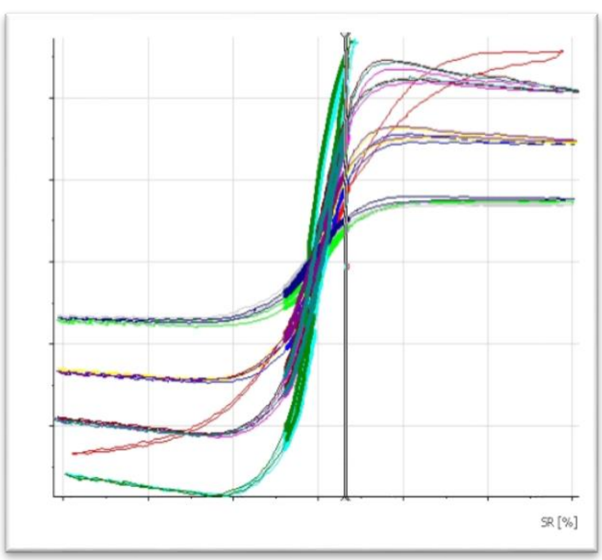
Description

Tractive force is the force in the X direction divided by the slip ratio and normalized by vertical load.

Set Up

The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.

Image

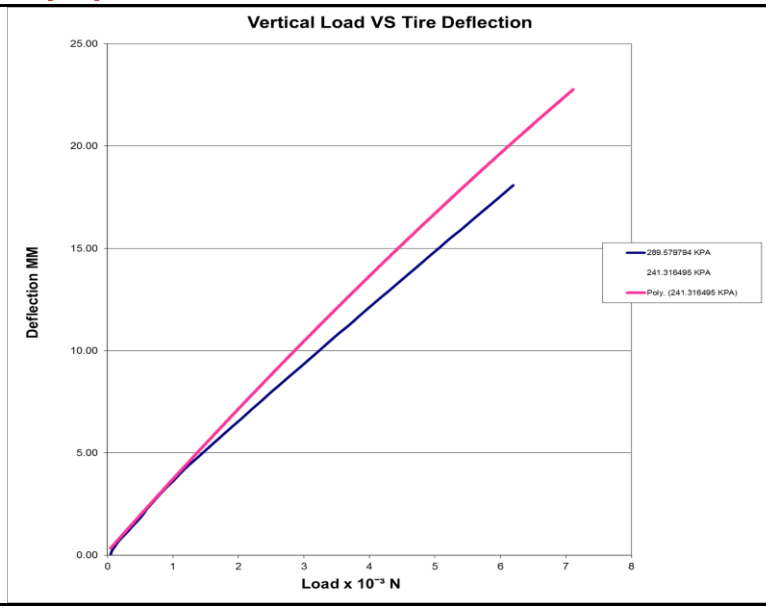


Criteria

To be confirmed

Report comparison to Reference Tire in 4.6

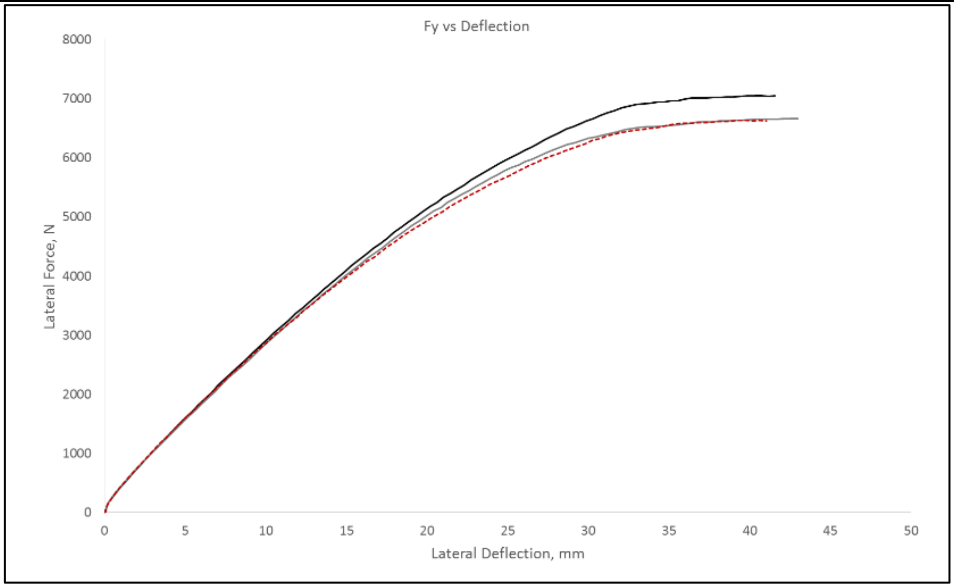
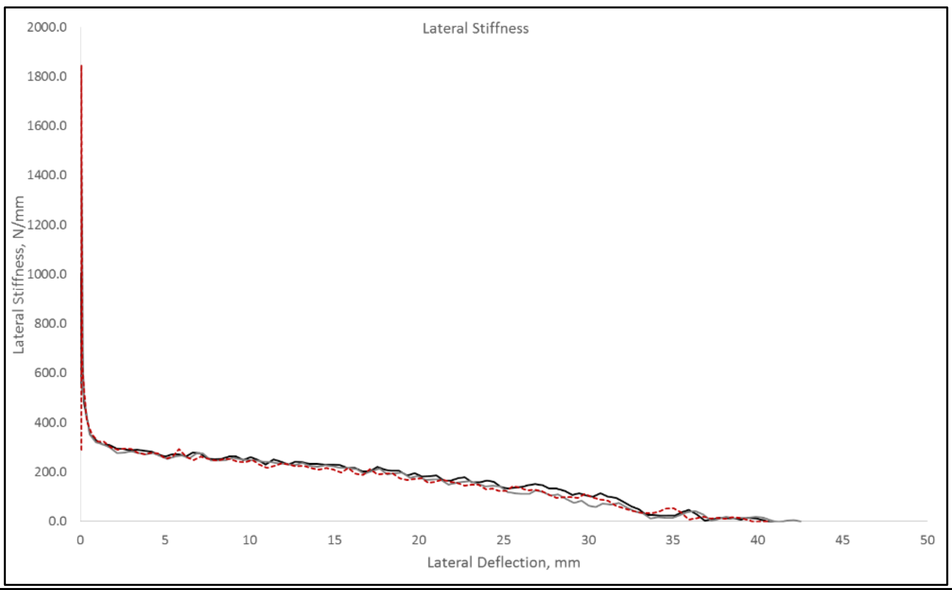
4.2.4. Vertical Stiffness

Description	Vertical stiffness is the slope of the vertical load verses tire radius plot.
Set Up	The output of the data should be in MF6.2 format with SWIFT, and adequate to create the below characterization plots.
Parameters	Pressure [kPa]: 276,290,310
Image	
Criteria	**305 N/mm ± 10% @ Application Pressure** Report comparison to Reference Tire in 4.6

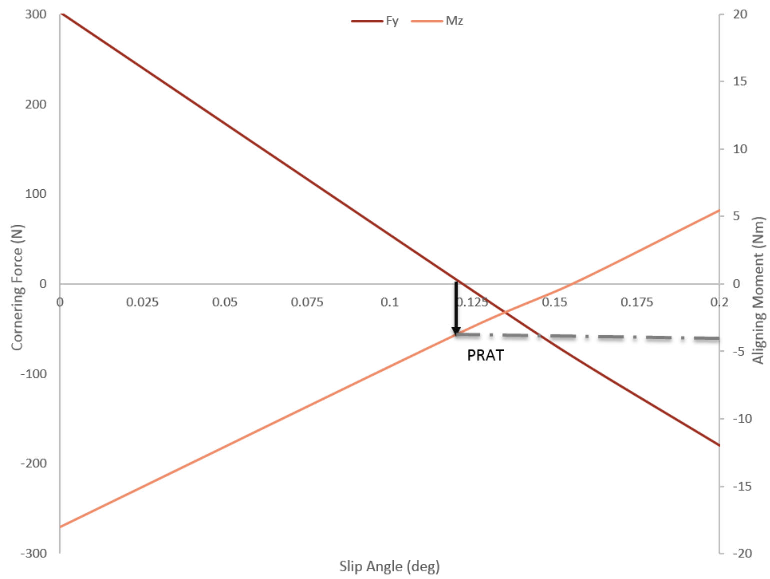
4.2.5. Vertical Damping

Description	Measure the vertical damping ratio of a free bouncing tire.
Set Up	Supplier Test Method. TBD.
Image	
Criteria	To be confirmed

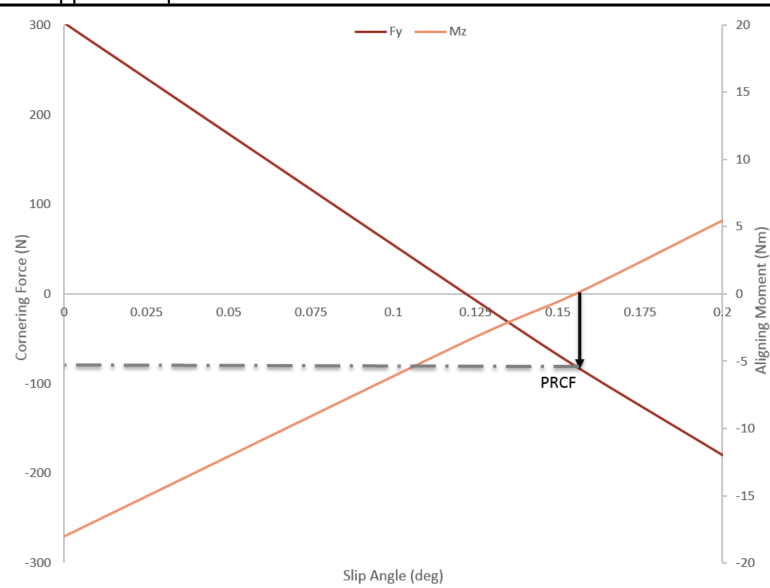
4.2.6. Lateral Stiffness

Description	Lateral stiffness is the slope of the lateral load verses tire deformation plot.
Set Up	The output of the data should be in MF6.2 format with SWIFT, and adequate to create the below characterization plots.
Parameters	Pressure [kPa]: 276,290,310
Image	
Criteria	Measured at Application Pressure: Lateral Stiffness ($K_{Y,1}$) at 5 mm deflection: **330 N/mm ± 5%** Lateral Stiffness at 20mm = **0.55 * $K_{Y,1}$ MIN** First derivative reported. Report comparison to Reference Tire in 4.6

4.2.7. Plysteer Residual Aligning Torque

Description	Plysteer Residual Aligning Torque is the moment about the Z-axis when lateral force is zero.
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.
Parameters	Load [kg]: 525,580 Pressure: Application pressure
Image	
Criteria	**2.0 ± 0.5 Nm**

4.2.8. Plysteer Residual Cornering Force

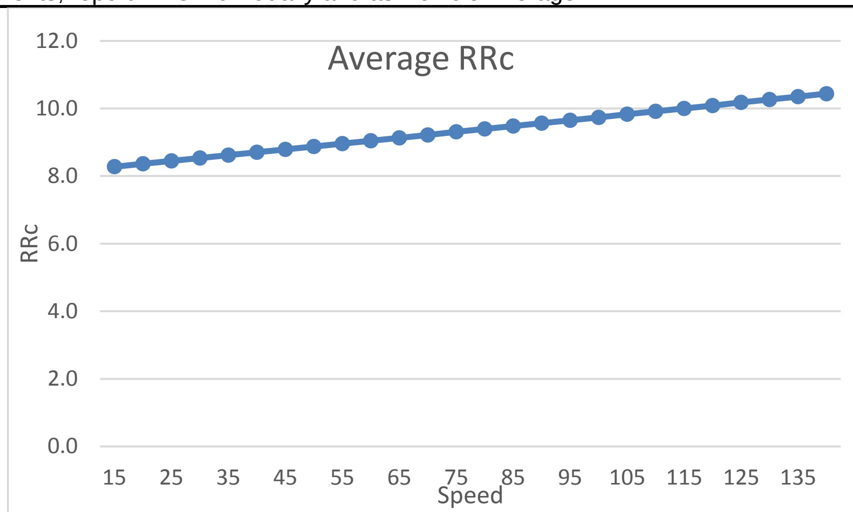
Description	Plysteer Residual Cornering Force is a force in the Y-Axis when Aligning Moment is 0.
Set Up	The output of the data should be in MF6.2 format, and adequate to create the below characterization plots.
Parameters	Load [kg]: 525,580 Pressure: Application pressure
Image	
Criteria	**75 N ± 20 %**

4.2.9. Relaxation Length (m/SA)

Description	Measure the relaxation length of the tire.
Set Up	Mathematically acquired from Cornering Stiffness and Lateral Stiffness $$L_R = \frac{C_\alpha}{K_Y} * \left(\frac{360\,°}{2\pi}\right)$$ $$mm = \frac{N/°}{N/mm} * \left(\frac{360\,°}{2\pi\ RAD}\right)$$
Image	
Criteria	R_L = **365 MAX** at Fz = 5.7 kN

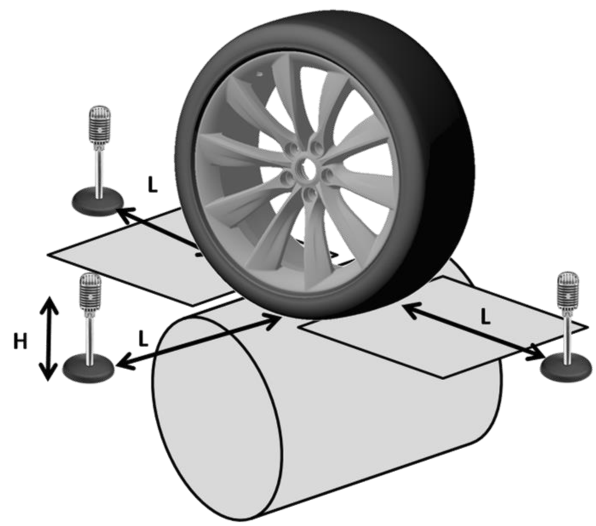
4.3. Rolling Resistance

4.3.1. Rolling Resistance – SAE

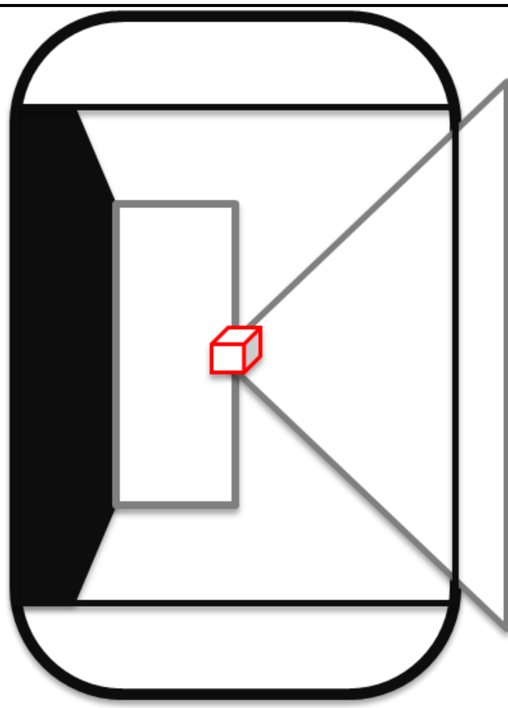
Description	Measure the effort required to keep a tire rolling.
Set Up	The rolling resistance coefficient should follow SAE J2452, with Tesla Parameter Modifications. $RR=F= P^{\alpha}L^{\beta}(a+bv+cv^2)$ RR Coefficient = 1000 (RR/L) Units: RR (N), L (N)
Parameters	**Pressure Sweep [kPa]:** 250,280,310 **Speed Sweep [kph]:** 115 - 15 **Load [% of Max Load Capacity]:** 70,85,100 **Samples:** 2 per S# construction. 4 additional per preferred construction For Split Fitments, report RRC individually and as Vehicle Average.
Image	
Criteria	**RR_C: 6.8 ± 0.20** Vehicle Average [Application Pressure, 80 kph,100% GVW load] Report: α,β,a,b,c coefficients for each tire tested Report comparison to Reference Tire in 4.6

4.4. Noise, Vibration, and Harshness

4.4.5. Sound Pressure Level

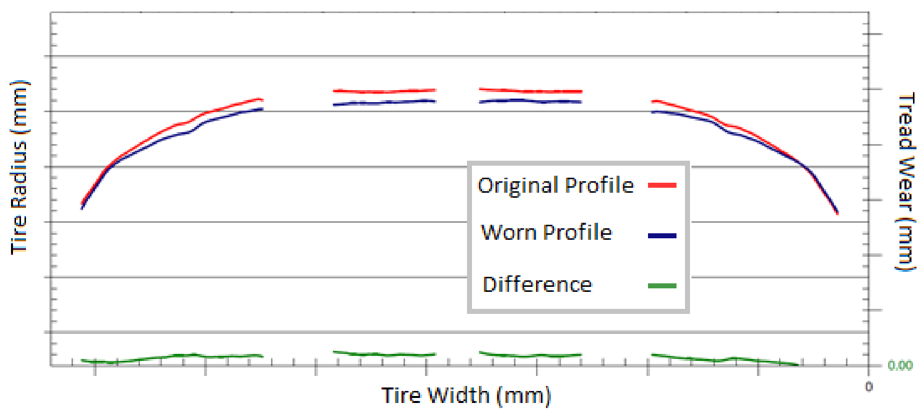
Description	Measure the tire road noise.
Set Up	a. In an acoustic chamber - measure the sound pressure level at the entrance, exit, and side a distance away from the tire as shown in the picture below for a duration of **10s.** Use **both smooth and rough surfaces** and at **50, 80, and 100 kph constant vehicle speeds**. b. Also, perform a **coast down from 120 to 25 kph** on **both smooth and rough surfaces** for **duration 60s.**
Parameters	**Load [kg]: 580** **Pressure [kPa]:** 276,**290** **L [m]: 1** **H [m]: 0.2**
Image	
Criteria	Data output FFT: 20 Hz to 20 kHz with a delta f of 4 Hz, Hanning Window, 50% overlap a. Color map (time-frequency), Linear weighting (non A-weighting) b. OAL verse time (non A-weighting) Report comparison to Reference Tire in 4.6

4.4.6. Structural Vibration

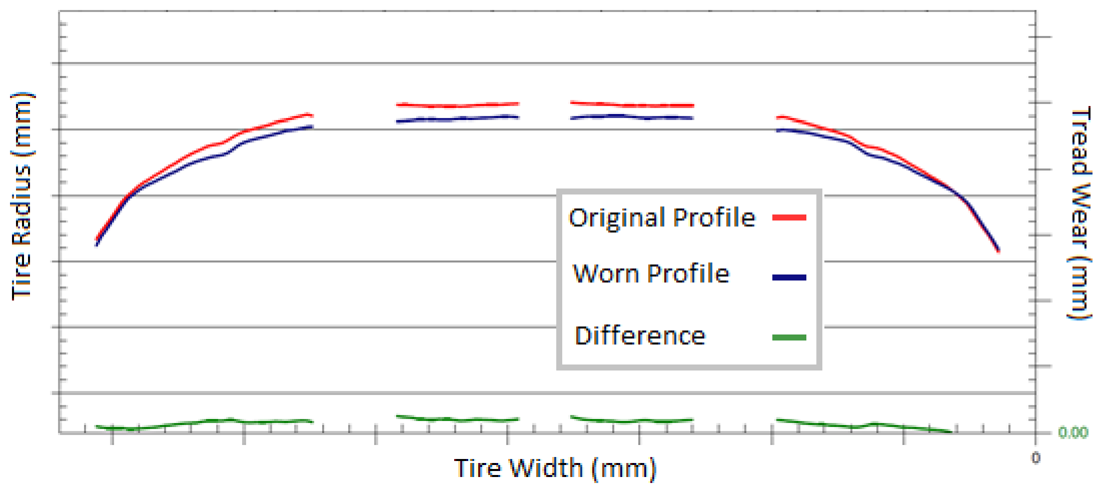
Description	Measure vibration of the bearing hub with a force transducer in each axis.
Set Up	a. In an acoustic chamber - measure the vibration of the bearing hub in each axis (X,Y,Z) for a duration of **10s.** Use **both smooth and rough surfaces** and at **50, 80, and 100 kph constant vehicle speeds**. b. Also, perform a **coast down from 120 to 25 kph** on **both smooth and rough surfaces** for **duration 60s.**
Parameters	**Load [kg]: 580** **Pressure [kPa]: 276,290**
Image	
Criteria	Force Transducer: FX, FY, FZ Data output FFT: 0 Hz to 400 Hz with a delta f of 0.5 Hz, Hanning Window, 50% overlap a. Color map (time-frequency) b. OAL verse time Report comparison to Reference Tire in 4.6

4.5. Tire Wear

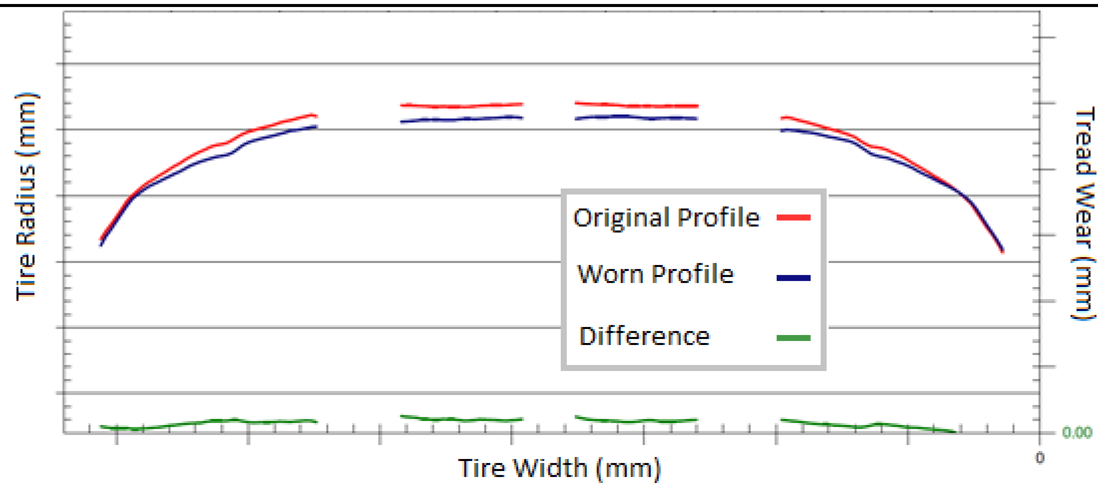
4.5.1. Fastest Wearing Groove

Description	Report the wear rate of the fastest wearing rib in mm per 10,000 miles.
Set Up	Supplier Test Method
Parameters	**Vehicle Camber**
Image	
Criteria	Worn Tire shall satisfy NVH Performance Criteria: 4.4.1 and 4.4.2 Fastest wearing groove shall be **≤15%** increase on Average Wear Rate 4.5.2

4.5.2. Average Wear Rate

Description	Report the average wear rate across the surface in mm per 10,000 miles
Set Up	Supplier Test Method
Parameters	**Vehicle Camber.**
Image	
Criteria	Worn Tire shall satisfy NVH Performance Criteria: 4.4.1 and 4.4.2

4.5.3. Irregular Wear Rate

Description	Report the wear rate across a rib that is different left to right of the tire center or across a single groove.
Set Up	Supplier Test Method
Parameters	**Vehicle Camber**
Image	
Criteria	Worn Tire shall satisfy NVH Performance Criteria: 4.4.1 and 4.4.2

4.6. Subjective Ride and Handling Rating

The subjective ride and handling rating should follow the industry standard 10 point rating system.

	Vehicle and Attribute Customer Rating System							90% customer satisfaction imperative		
Engineering Rating index	1	2	3	4	5	6	7	8	9	10
Evaluation of attribute performance	Not Acceptable		Poor		Borderline	Acceptable	Fair	Good	Very Good	Excellent
Customer satisfaction(for the attribute)	Very dissatisfied				Somewhat Dissatisfied	Fairly well satisfied		Very Satisfied	Completely satisfied	
Improvement desired by	All Customers			Average customer		Critical customer		Trained observer		Not perceptible

Figure 3: Sample Spider Chart. Chart shall include all Subjective Criteria from 4.6.1 as compared to a Reference Tire.

Current Reference for S0: Continental ProContact RX T0 (235/40R19)	Reference	TARGET	Ranking	Target Construction
Steering				
On-Center Response & Feedback				
Off-Center Response & Feedback				
Steering Linearity, Gain, & Precision				
Phase Lag				
Ride				
Primary Ride				
Secondary Ride				
Wet Handling				
Grip Level				
Sliding Grip				
Wet +/- 0.3 g Linearity, Progressivity				
Grip Recovery				
Wet Braking Traction				
Throttle On/Off Reaction				
Braking in a Turn				
Balance				
Dry Handling				
Grip Level				
Understeer/Oversteer Balance				
Lift Off in a Turn				
Straight-Line Tracking Stability				
Yaw Stability				
Lateral Buildup				
Dry +/- 0.5 g Linearity, Progressivity				
Dry Transient Handling				
Dry Limit Handling Potential				
Dry Braking				

Subjective targets to be considered at Mule Development

4.7. Uniformity

Description	Per SAE J332, Test 4.7.1 thru 4.7.6
Parameters	Rim Width: Per Axle Load: GVW Per 2.3 Inflation Pressure: Per 2.3

Item	Description	Criteria	Markings
4.7.1	Conicity	±70 N	1 stripe across the tread for the negative class 2 stripes across the tread for the positive class
4.7.2	Radial Run-out	± 1 mm	
4.7.3	Radial Force Peak to Peak (RFPP)	105 N	
4.7.4	Radial First Harmonic (RF1H)	75 N	Red paint dot at the high point
4.7.5	Lateral Force Peak to Peak (LFPP)	80 N	
4.7.6	Lateral Run Out	± 1 mm	

5. Market Countries

Zone	Country
RENA	United States
RENA	Canada
REAP	South Korea

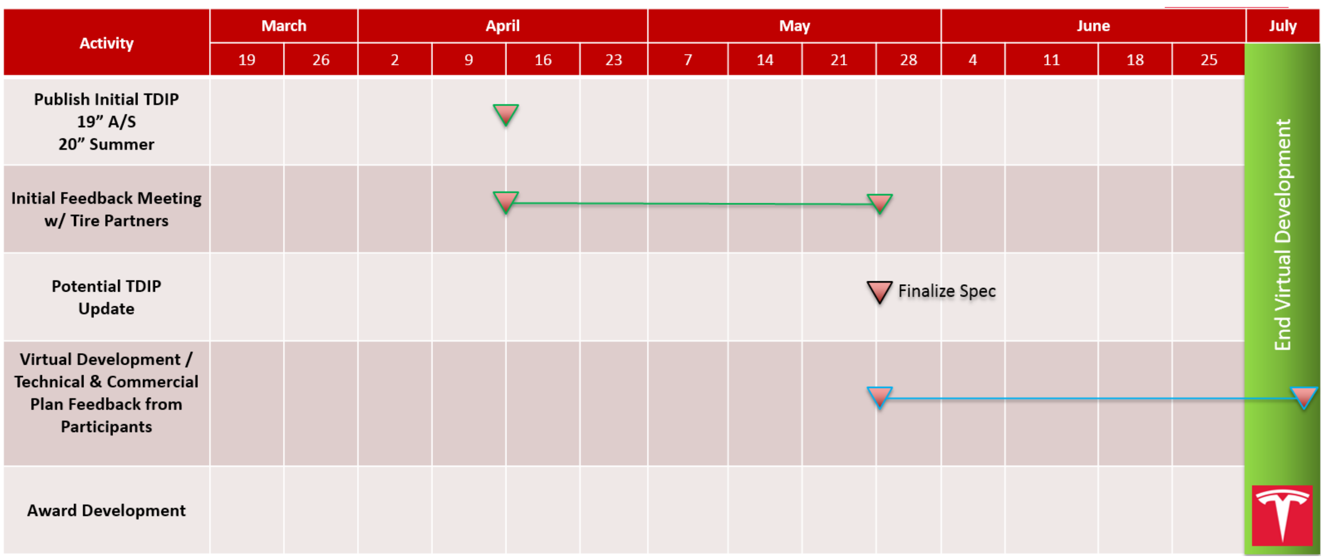
6. Appendix A: Referenced Procedures and Specifications

The following documents were referenced in the creation of this document. If there is a need for clarification or detail beyond the scope of this document, the following documents shall be used as a guideline for such issues.

- *SAE* J2047 - *Tire Performance Terminology*
- SAE J2452 - *Stepwise Coastdown Methodology for Measuring Tire Rolling Resistance*
- SAE J332 - *Testing Machine for Measuring the Uniformity of Passenger Car and Light Truck Tires*
- ISO 362 - *Pass-by noise testing standard*

7. Appendix B: Program Timing

Projected milestones, including submissions, vehicle builds, and validation requirements. Initial timing subject to change pending program schedule updates.



Virtual development through June and July 2018
Mule development (Sub 0) Sept. 2018
Joint Development (Sub 1) Jan 2019
Joint Development (Sub 2) May 2019